

Effector B-cell model: post-meeting update

From threshold sweeps to basin structure and formal saddle-node continuation

Main take-away

b6 creates a saddle-node tumor-control tipping point; b5 shifts the required b6 threshold.

Continuation confirms stable high- and low-tumor equilibria.

Basin maps show initial tumor/NK/B context controls final fate.

Two-parameter fold continuation links MDSC suppression to the b6 threshold.

The analysis follows Prof. MacLean's meeting question

Question: is the b6 threshold a true dynamical transition or only an apparent simulation threshold?

1

Up/down continuation

tests history dependence

2

Formal continuation in b6

finds equilibrium branches

3

Basin maps + trajectories

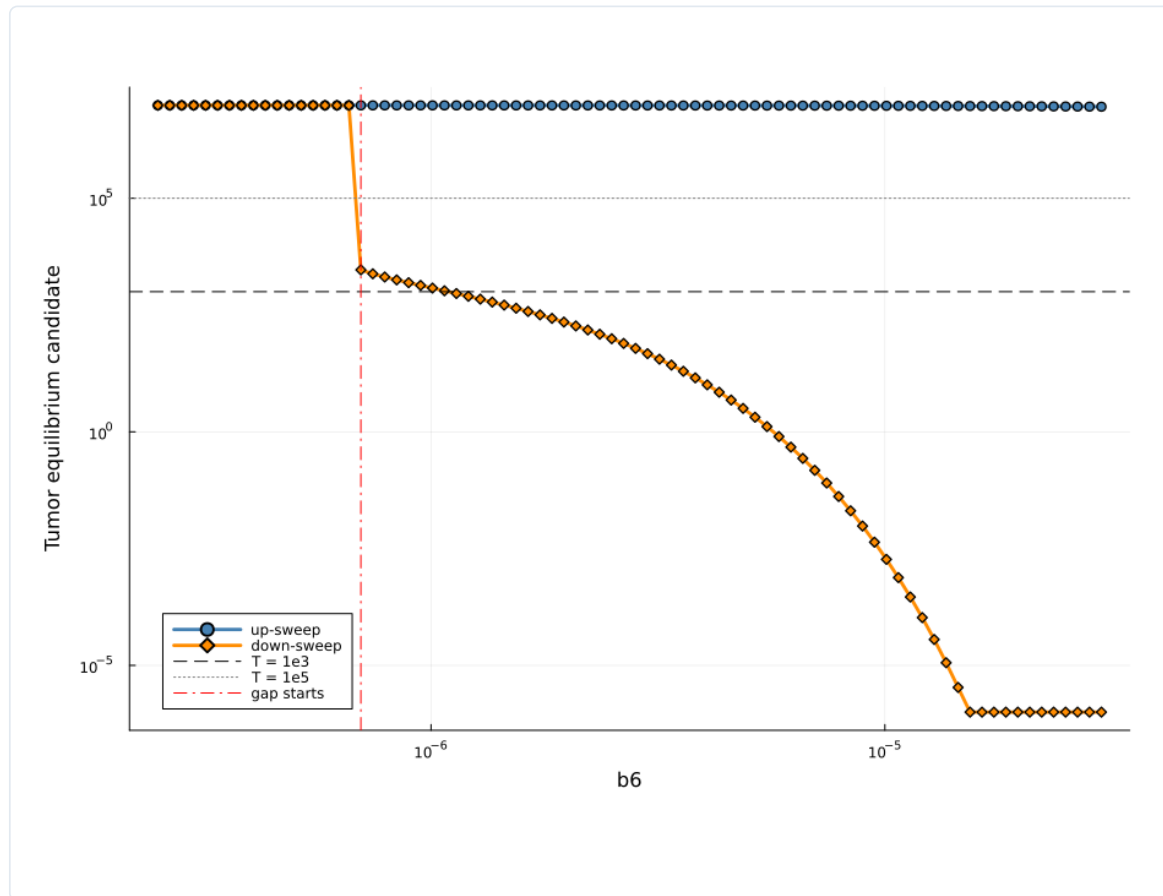
tests which initial states reach each attractor

4

b5–b6 fold continuation

tests how MDSC suppression shifts the tipping point

Up/down continuation reveals history dependence

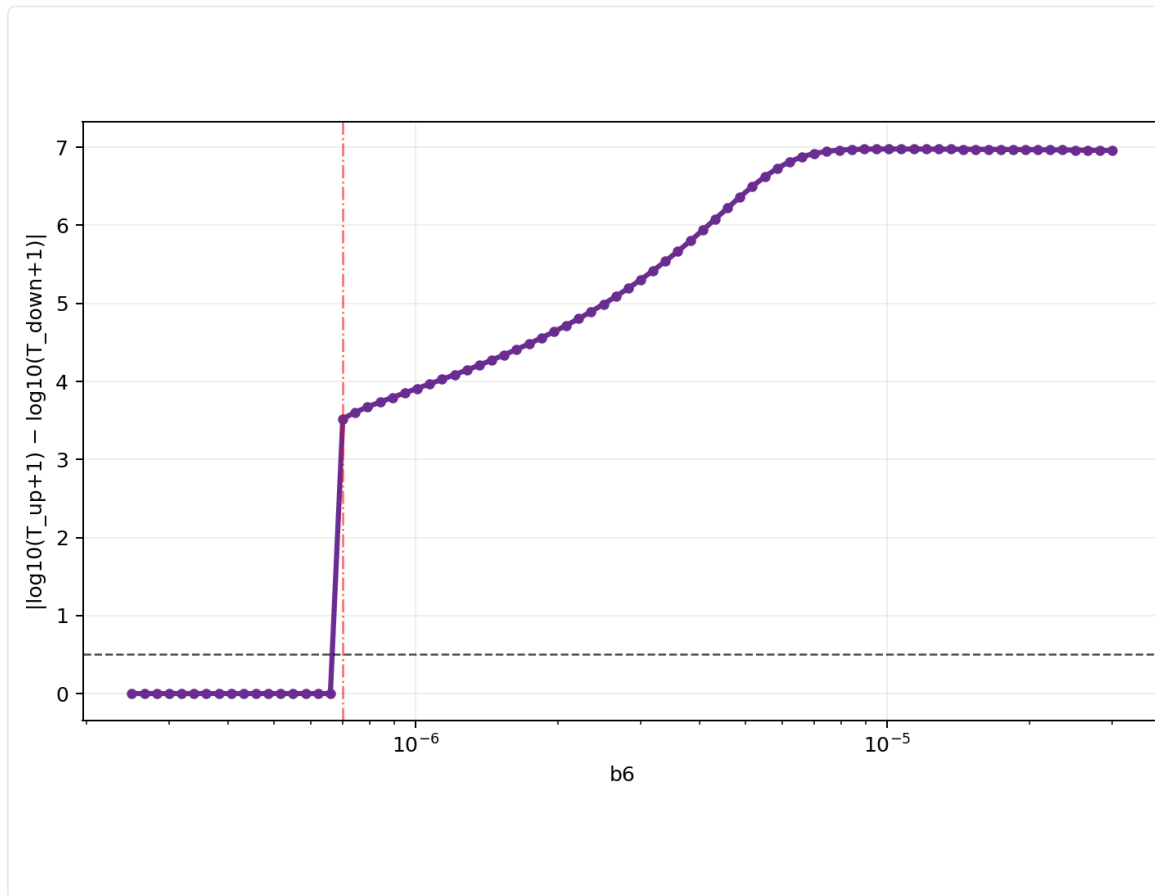


Up-sweep remains on the high-tumor branch.

Down-sweep follows a lower-tumor branch after the threshold.

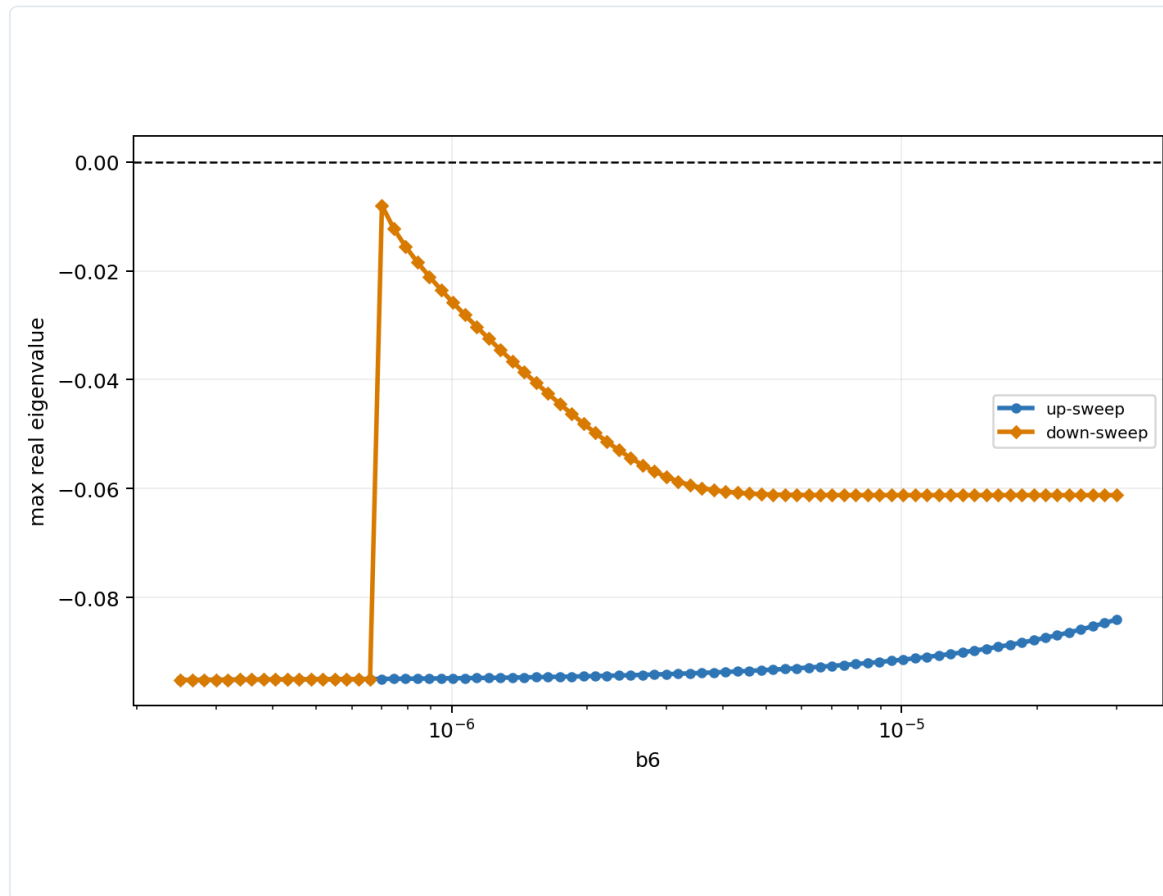
Same parameter range can support different long-time outcomes.

The hysteresis gap opens near the b6 threshold



The up/down branches agree at low $b6$.
A large log-scale gap appears near $\sim 7 \times 10^{-7}$.
This is the first numerical evidence for path dependence.

Residuals and stability support true branch candidates

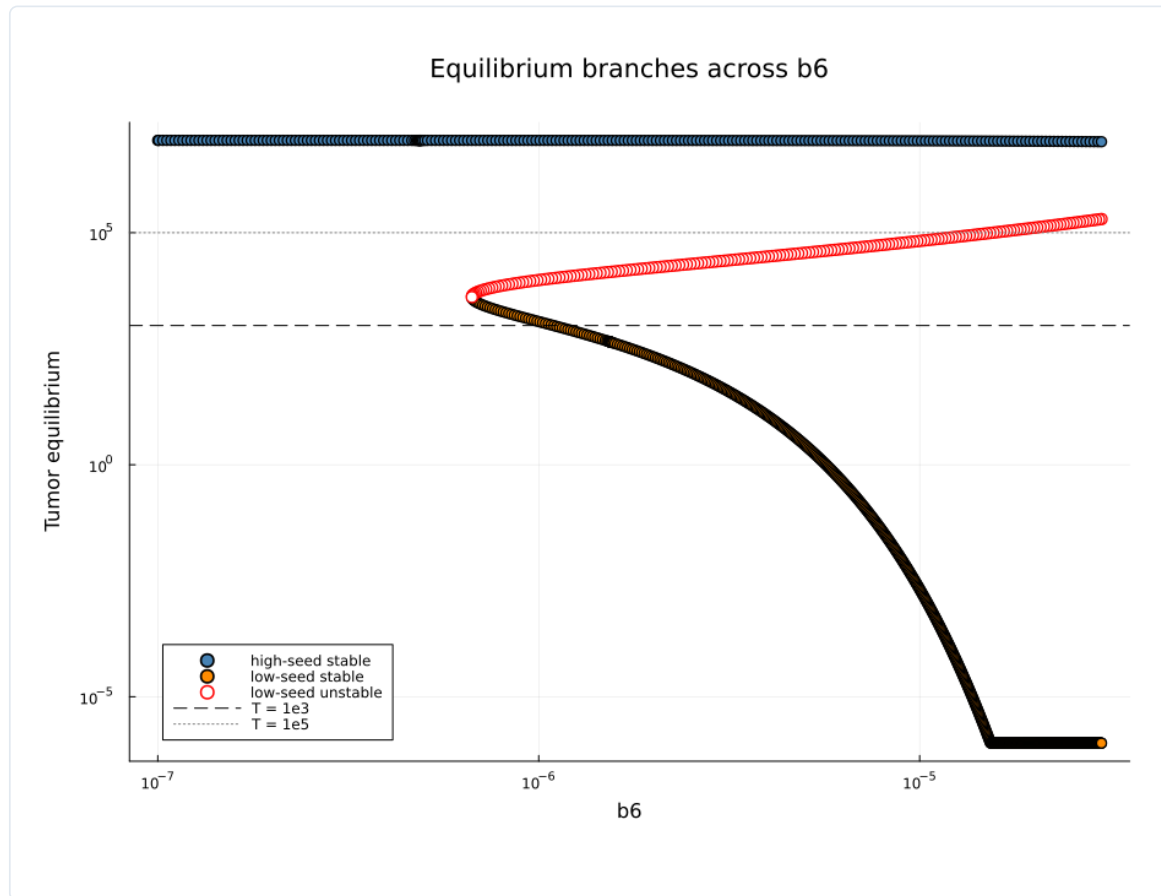


Max real eigenvalues stay negative on candidate stable branches.

Residual checks confirm the states are near equilibria.

This addresses the concern that flat trajectories may be slow transients.

Formal continuation traces high- and low-tumor equilibria

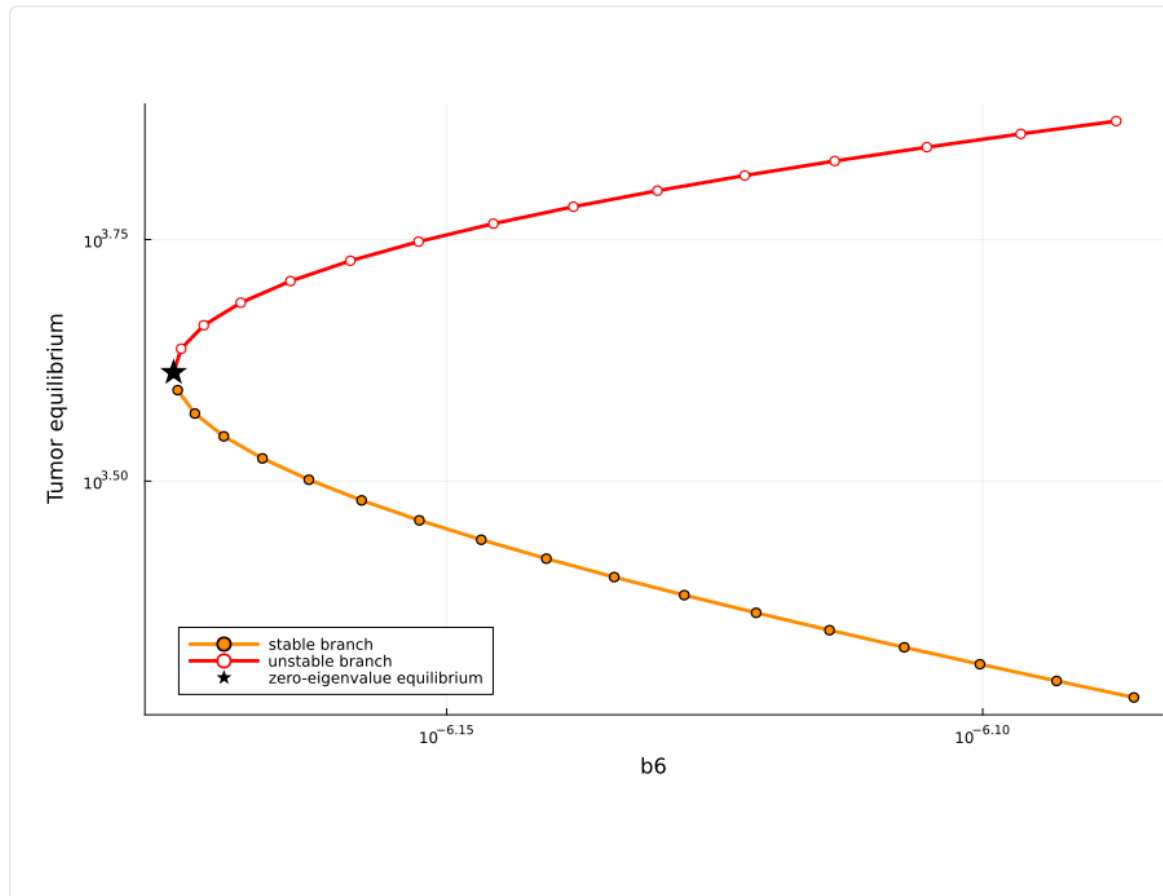


Continuation follows equilibria directly, not only trajectories.

High- and low-tumor branches coexist over overlapping b_6 values.

This upgrades the threshold observation into equilibrium structure.

Stable and unstable tumor equilibria meet at a saddle-node

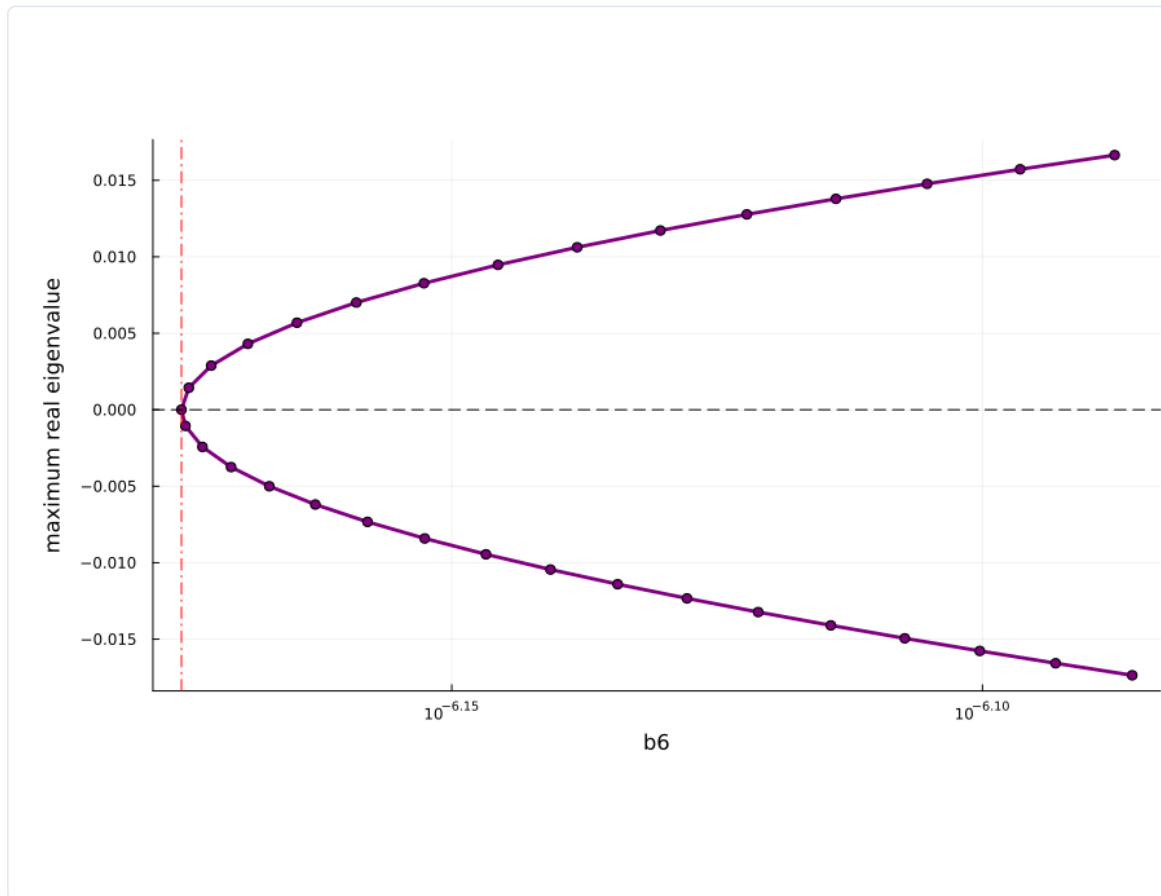


A stable low-tumor branch meets an unstable branch.

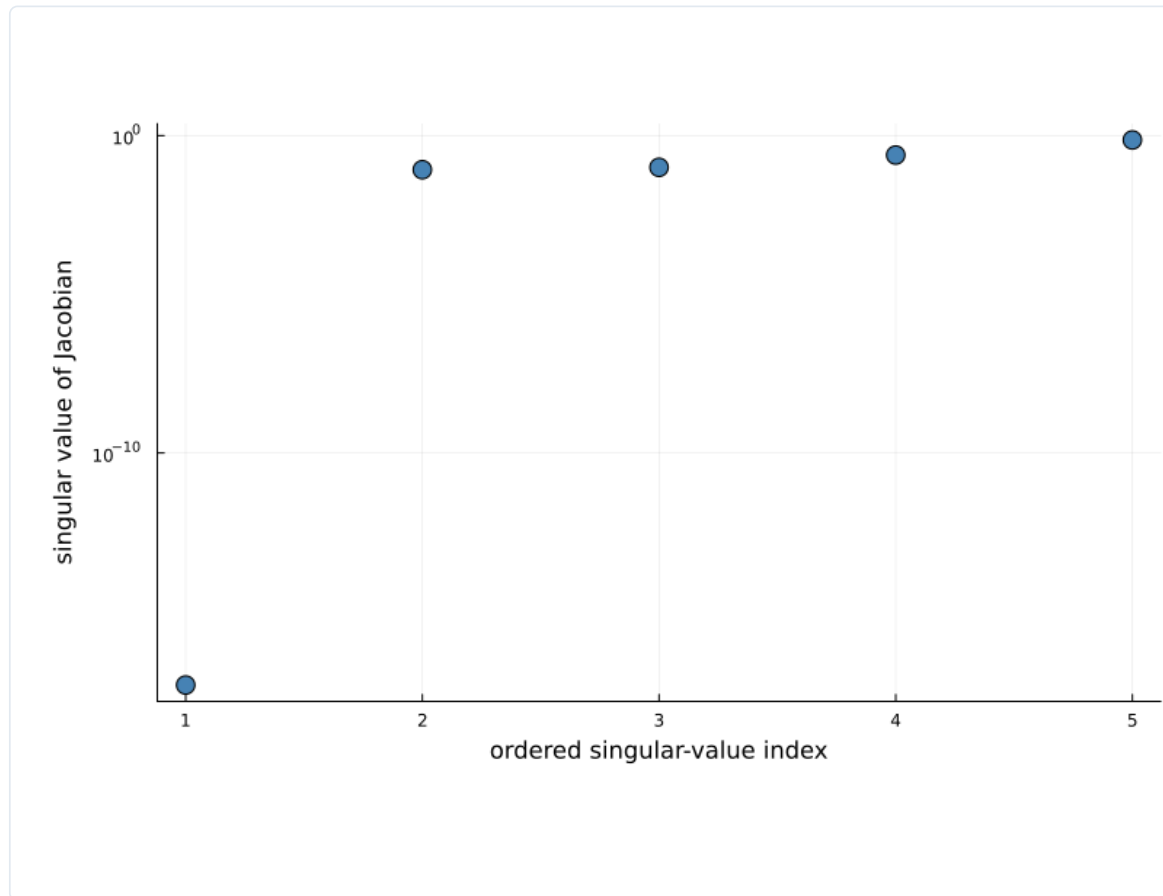
The critical point occurs near $b_6 \approx 6.68 \times 10^{-7}$.

This explains why the model shows a tipping-point response.

A simple real eigenvalue crosses zero at critical b6



The Jacobian has one isolated zero mode



Only one singular value is near zero.
The remaining modes are far from zero.
This supports a simple codimension-one
bifurcation, not a degenerate point.

Final regimes are threshold-based long-time classifications

controlled

$T(10000) < 1e3$

intermediate

$1e3 \leq T(10000) < 1e5$

tumor-dominant

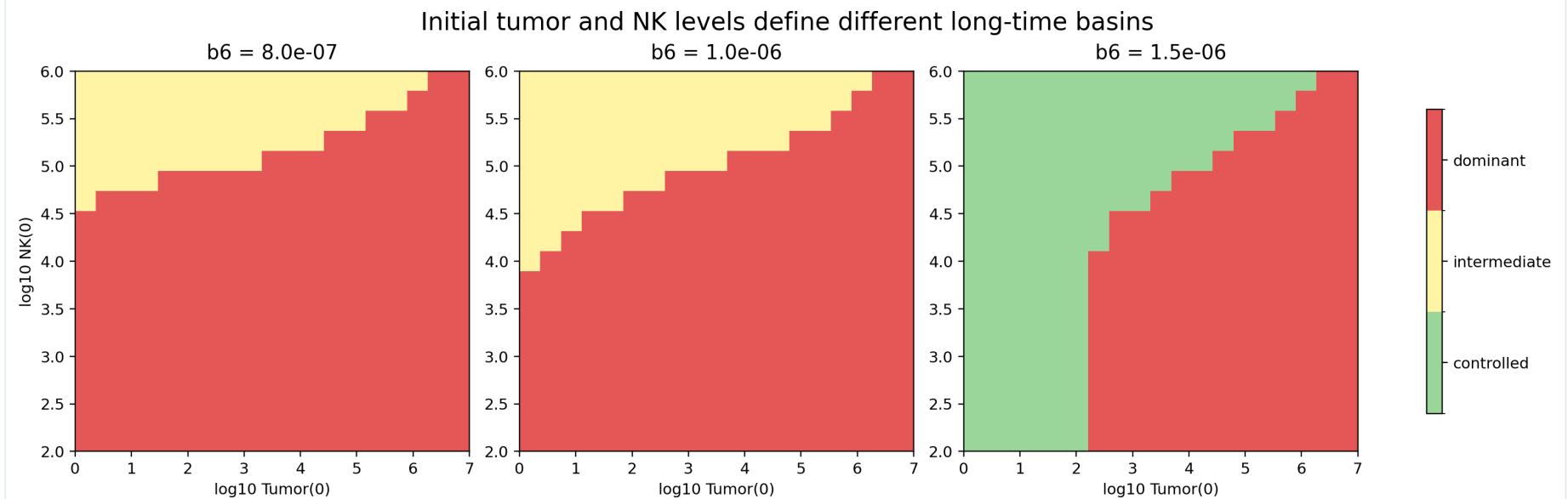
$T(10000) \geq 1e5$

The basin maps classify the final numerical outcome at day 10,000.

The intermediate label is value-based; it does not by itself prove a separate stable equilibrium branch.

Equilibrium stability is assessed separately using continuation and Jacobian eigenvalues.

Higher b_6 expands the tumor-control basin

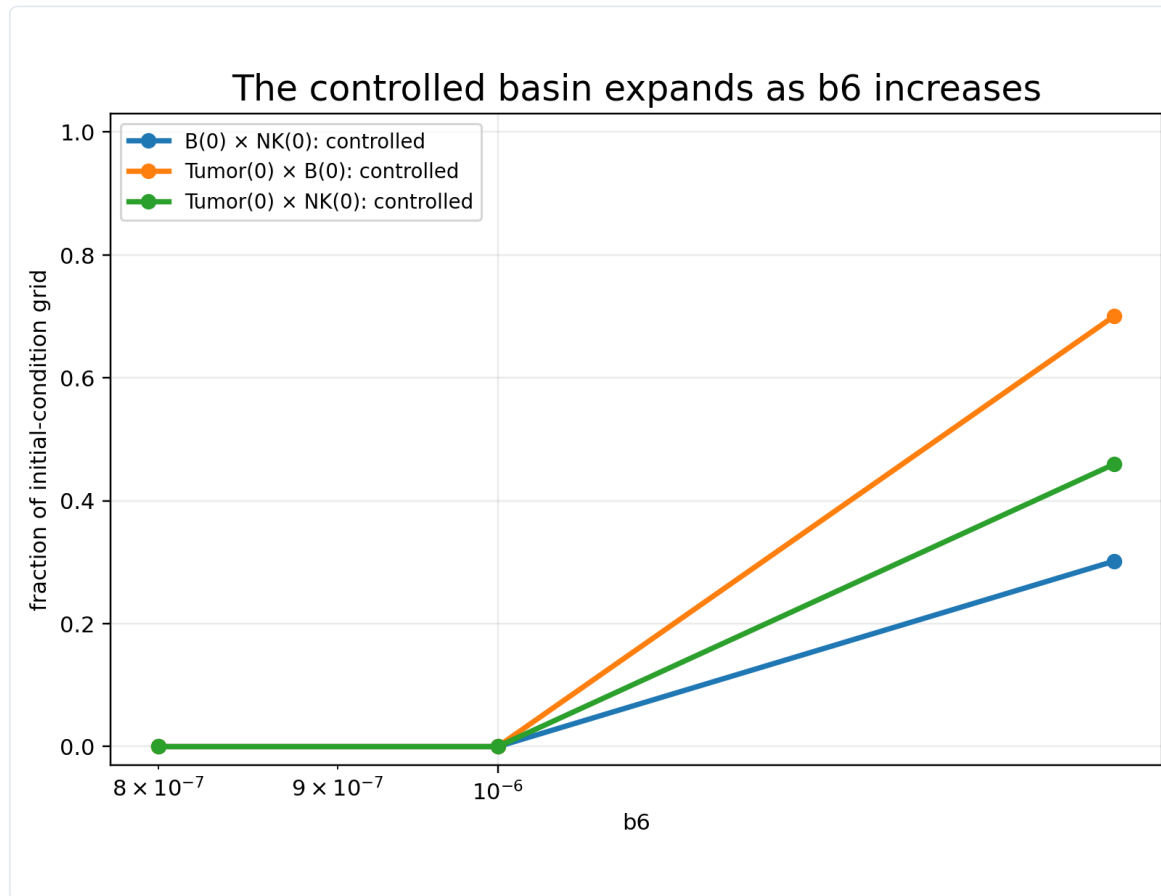


Basin maps vary $\text{Tumor}(0)$ and $\text{NK}(0)$ at fixed b_6 .

Final-state criteria: controlled $T(10000) < 1e3$; intermediate $1e3 \leq T(10000) < 1e5$; dominant $\geq 1e5$.

Intermediate is a value-based long-time outcome, not a separately proven stable branch.

The controlled basin grows as b_6 increases

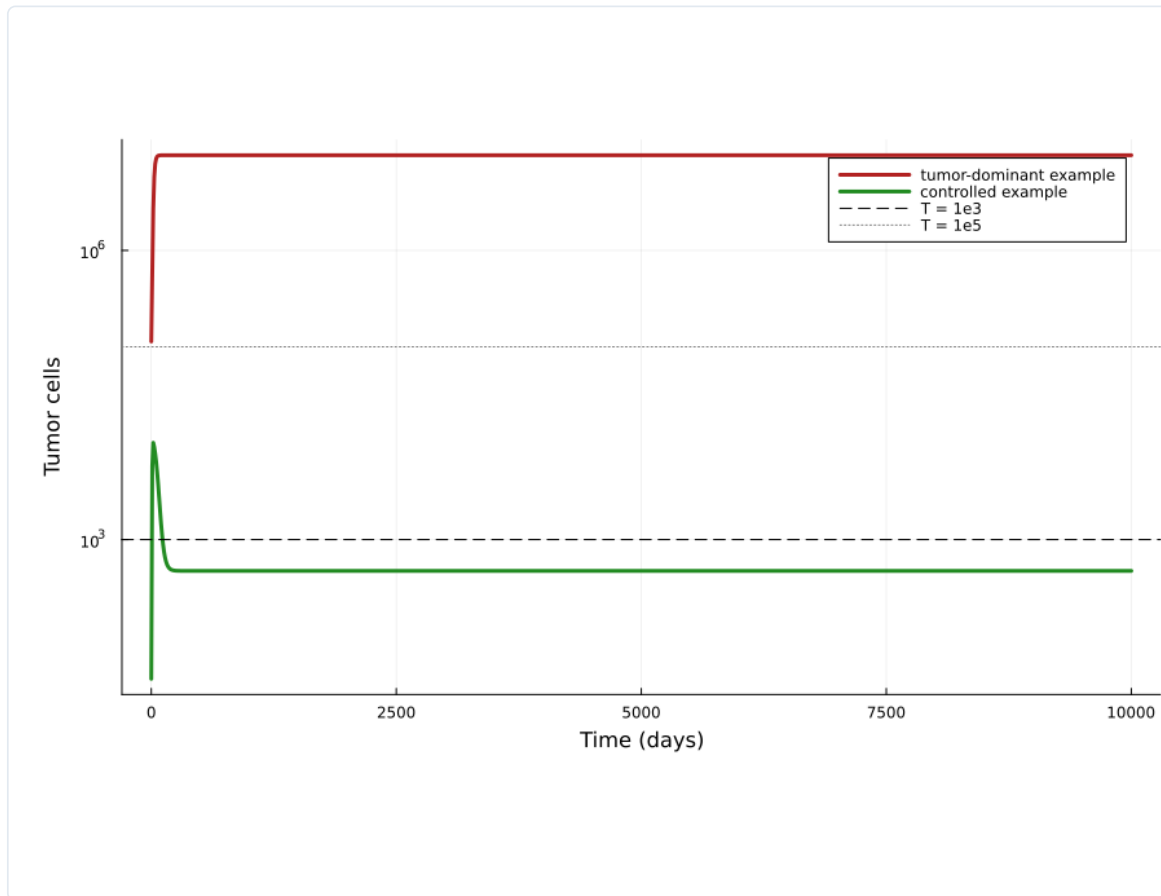


Controlled basin fraction is zero at lower b_6 values.

A controlled basin appears at $b_6 = 1.5 \times 10^{-6}$.

Tumor/B and tumor/NK maps show the largest control regions.

The same b6 can produce different tumor trajectories

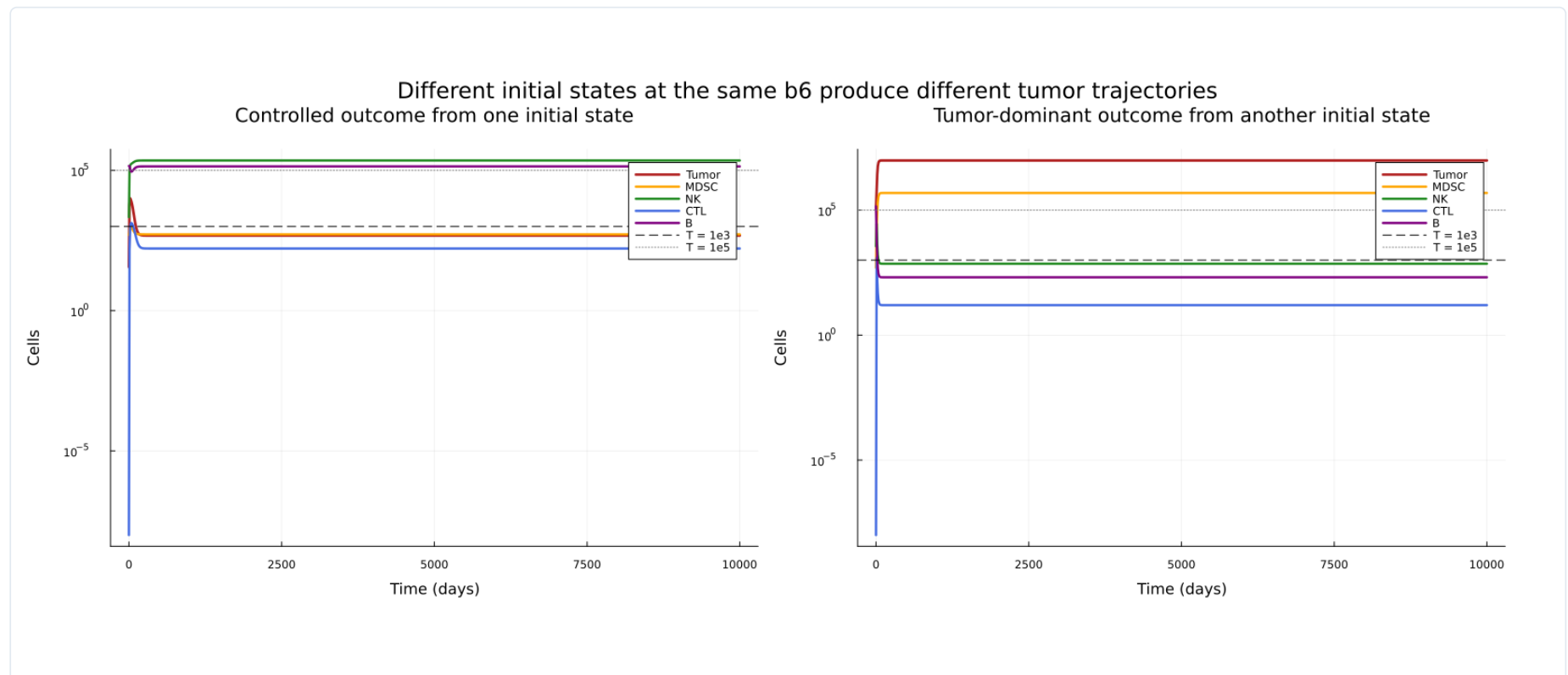


Two initial states were simulated with the same b6.

One trajectory remains tumor-dominant; the other becomes controlled.

This is a trajectory-level view of different basins of attraction.

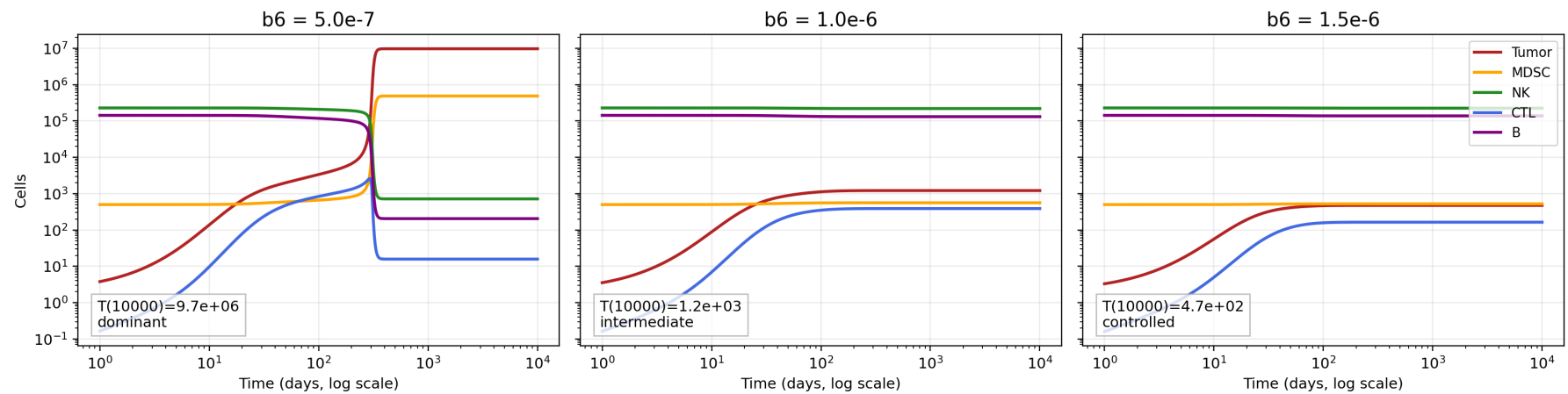
Full-state trajectories show two different long-time regimes



Tumor, MDSC, NK, CTL, and B dynamics separate by initial state.
The controlled case maintains low tumor with different immune dynamics.
This is useful because Prof. MacLean specifically values trajectories.

Same initial condition, different b6 values change the trajectory

Same initial condition with different b6 values produces different long-time dynamics

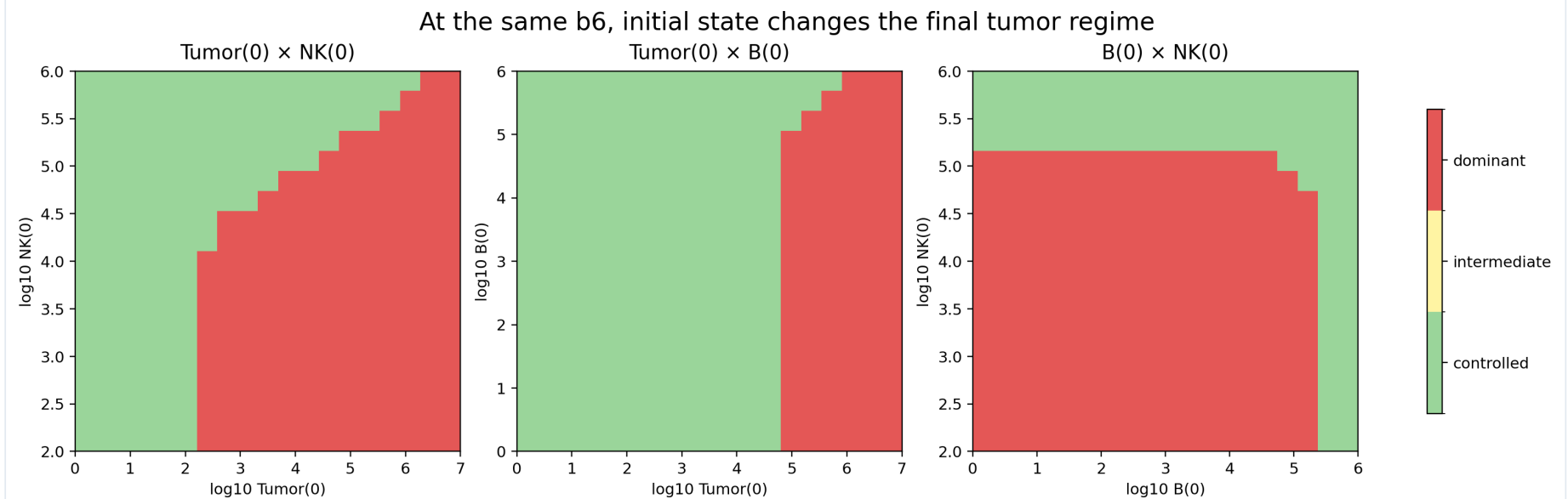


Same baseline IC and model structure; only b_6 changes.

This connects the final-state criteria to full Tumor/MDSC/NK/CTL/B trajectories.

Increasing b_6 shifts the trajectory from tumor-dominant/intermediate behavior toward controlled behavior.

Initial immune context changes long-time tumor fate

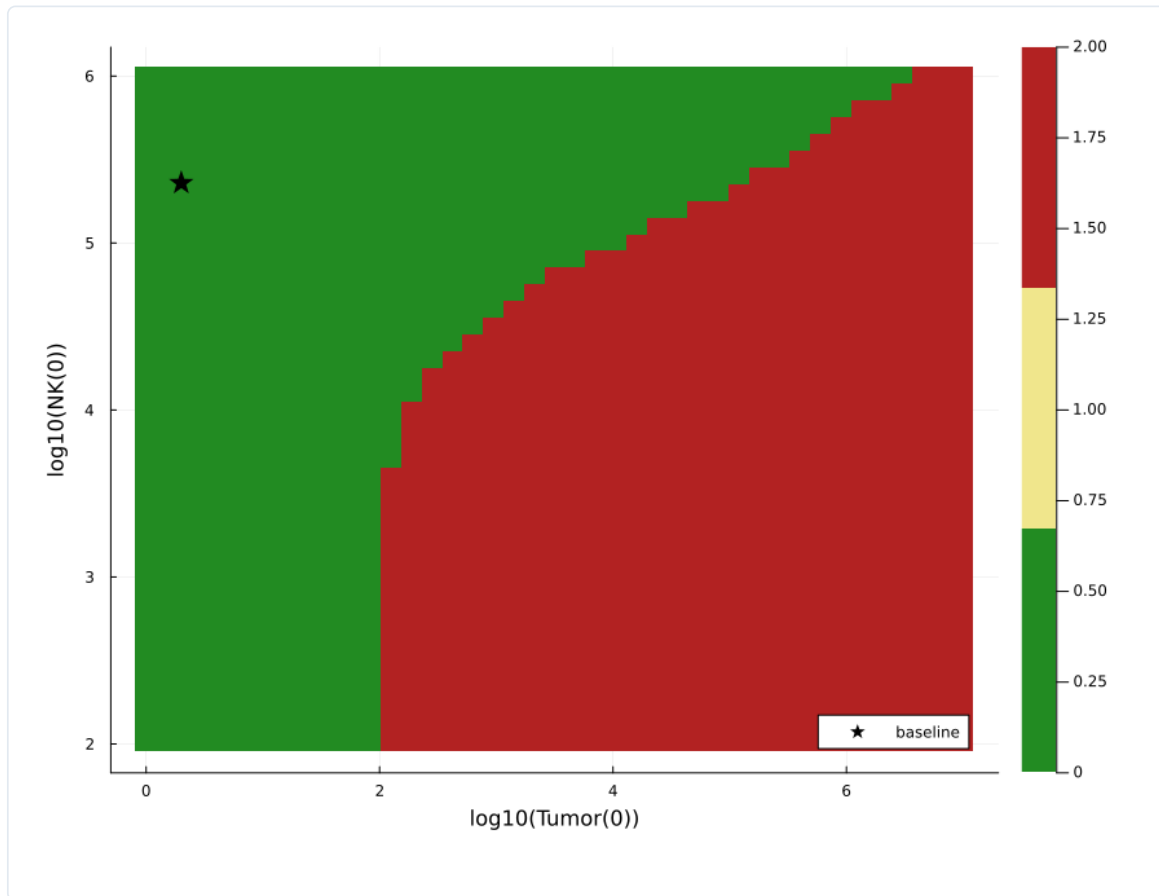


At $b_6 = 1.5 \times 10^{-6}$, different IC pairs produce different regimes.

For the $\text{B}(0) \times \text{NK}(0)$ panel, $\text{Tumor}(0)$ was fixed at $1e4$ cells.

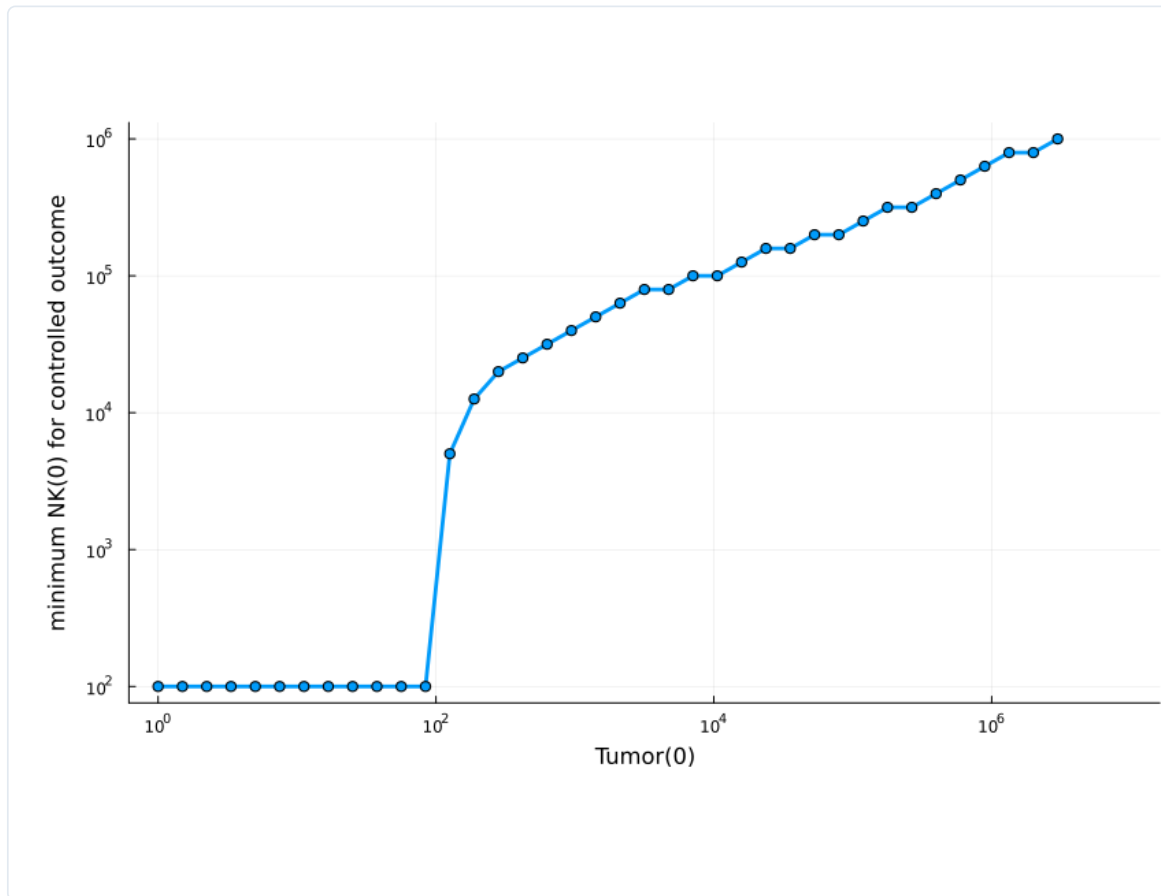
This makes the immune-context map harder to control than maps with lower tumor burden.

Tumor and NK initial levels separate long-time regimes



A refined $\text{Tumor}(0) \times \text{NK}(0)$ map sharpens the basin boundary.
Low tumor and high NK favor the controlled state.
High tumor or low NK favor tumor-dominant outcomes.

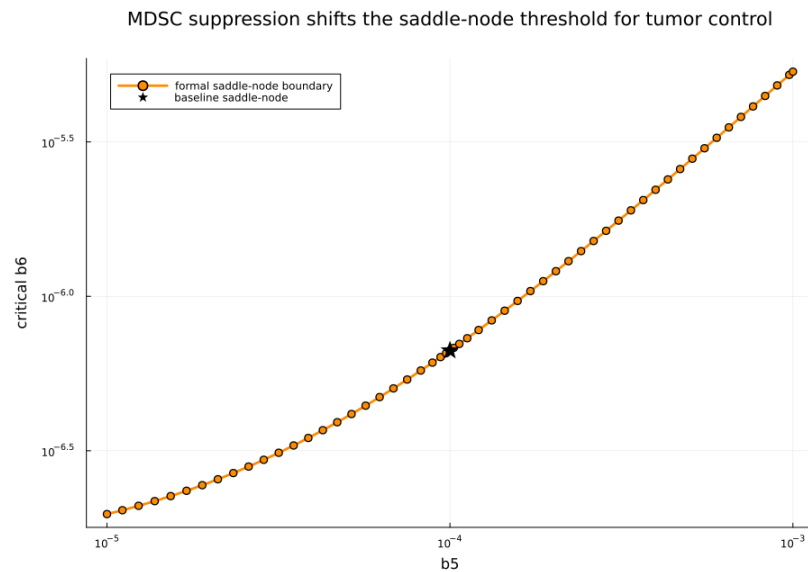
More initial tumor requires more NK for control



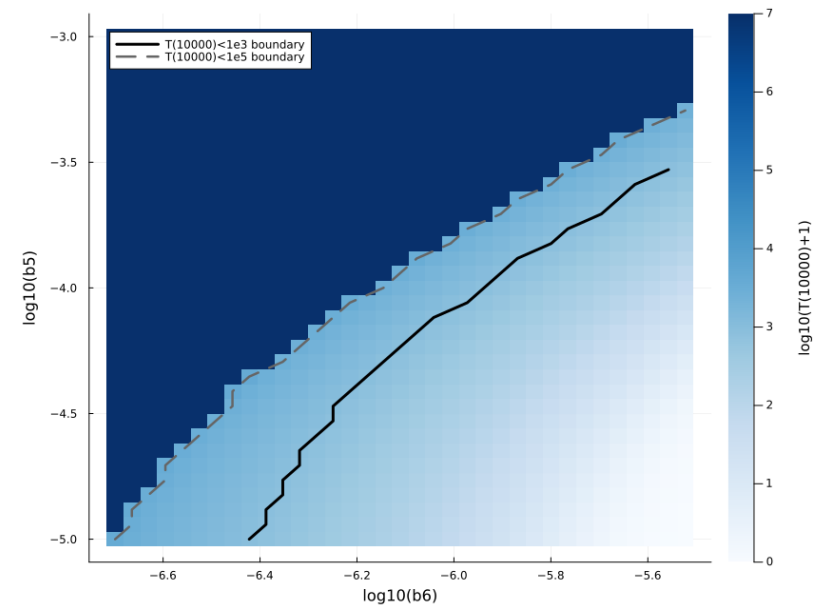
The boundary summarizes the refined Tumor(0) $\times$ NK(0) map.

As initial tumor increases, the minimum NK(0) for control rises.

MDSC suppression shifts the saddle-node threshold for tumor control



Formal fold curve: higher b_5 requires higher critical b_6

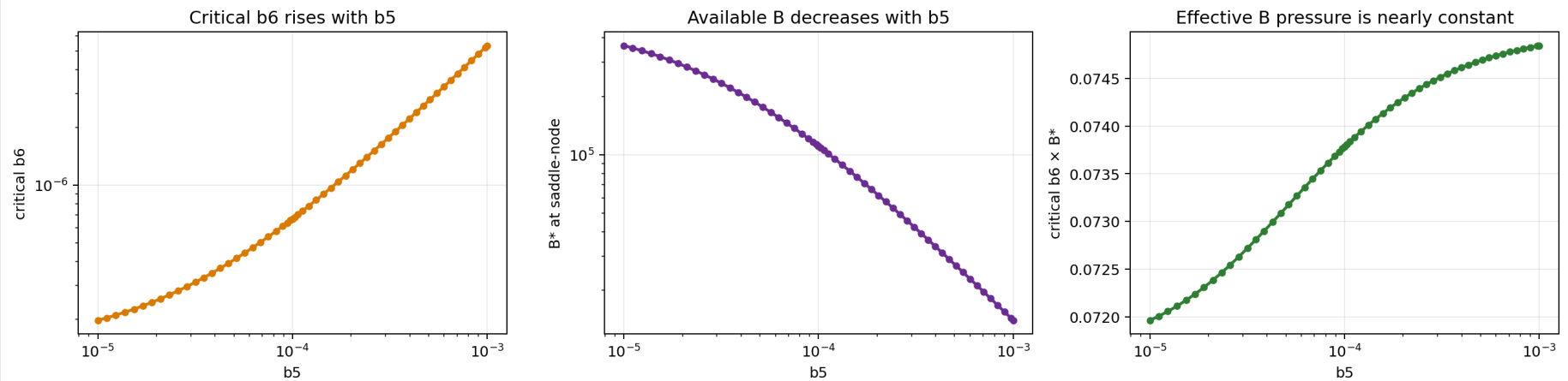


Heatmap reference: same trend appears in finite-time tumor-burden simulations

MDSC suppression shifts both the bifurcation boundary and the practical control boundary

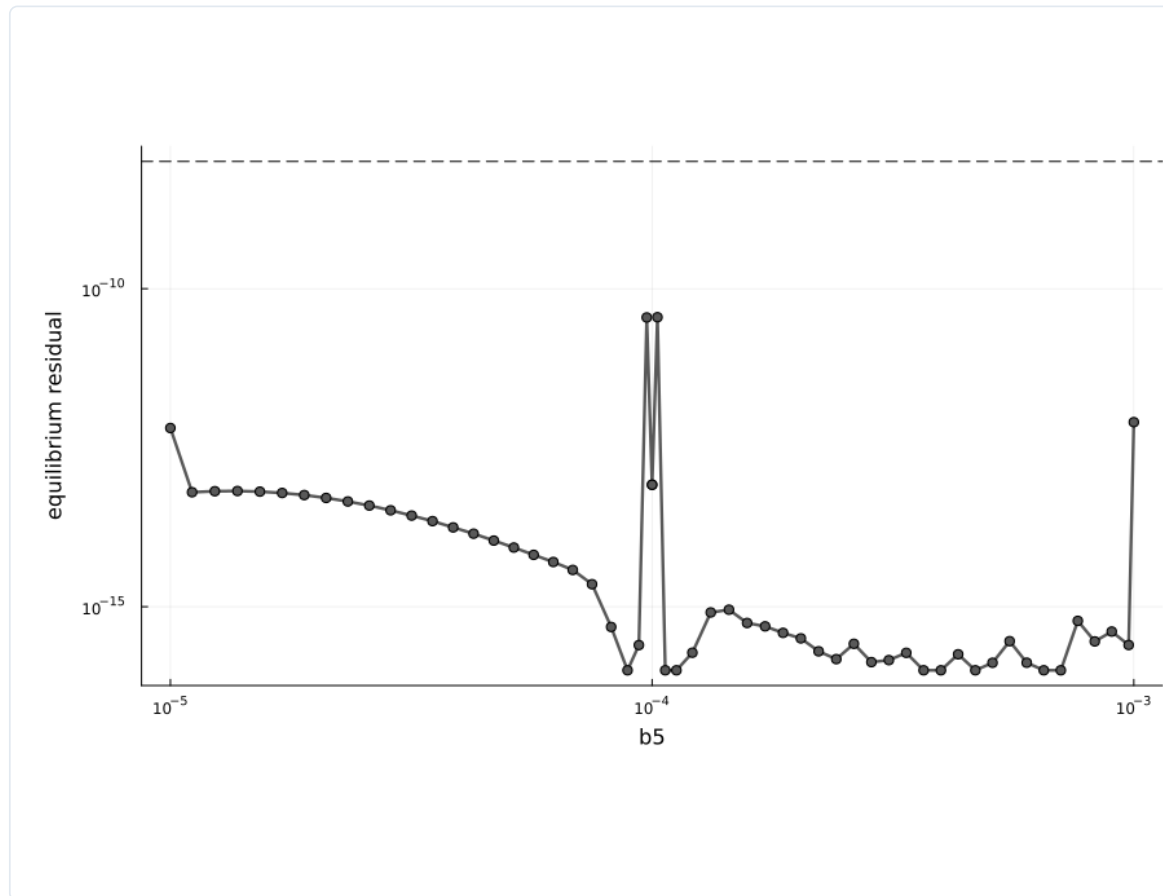
MDSC suppression is offset by stronger B-cell potency

MDSC suppression is offset by stronger B-cell anti-tumor potency



B^* decreases as b_5 increases.
Critical b_6 rises to compensate.
The product critical $b_6 \times B^*$ stays nearly constant.

The fold curve is numerically accurate



Residuals remain far below the numerical tolerance line.

The curve satisfies the steady-state equations.

This supports using the curve as a formal bifurcation boundary.

Summary: the model supports a tumor-control tipping-point mechanism

Main conclusions

b6 creates a saddle-node tumor-control threshold.
Stable high- and low-tumor outcomes coexist over a b6 range
Initial tumor/NK/B context determines which basin is reached.
b5 shifts the formal saddle-node boundary upward.

Questions / next directions

Should we repeat fold continuation under rare-B or NK-poor settings?

Should focused SDE simulations be run near the deterministic saddle-node?

effector B-cell activity creates a control attractor, while MDSC suppression and initial immune context determine whether the system can access it